

Dart Programming Language

String :-

Letter / Digit / Special Character
Punctuation Marks / Emoji

```
void main() {
```

```
String text1 = "Flutter"; // Double Quote
```

```
String text2 = 'Dart'; // Single Quote
```

```
Print(text1); // Flutter
```

```
Print(text2); // Dart
```

```
}
```

Multi-Line String

```
void main() {
```

```
String text = '''
```

```
Welcome to Dart
```

```
This is multi Line String.
```

```
''';
```

```
Print(text);
```

```
}
```

Empty String :-

```
String text = "";
```

length

```
void main() {
```

```
String text = "Dart";
```

```
Print(text.length); // 4
```

```
}
```

How to store a String in memory?

```
void main() {
```

```
String str = 'Dart';
```

```

    print(str); // Dart
}

```

UTF → Unicode Transformation Format

A-Z (65 - 90)

a-z (97 - 122)

D → 68 ✓

a → 97 ✓

z → 114 ✓

t → 116 ✓

(01)

print(str.codeUnits)

	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
	128	64	32	16	8	4	2	1
68		1	0	0	0	1	0	0
97		1	1	0	0	0	0	1
114		1	1	1	0	0	1	0
116		1	1	1	0	1	0	0

68 → 000 000 000 1000 100 ✓ → D
 97 → 000 000 000 110000 1 ✓ → a
 114 → 000 000 000 11100 10 ✓ → z
 116 → 000 000 000 1110100 ✓ → t

Emoji ← Unicode:

Print(emoji.codeUnits); // [55357, 56832]

Print(emoji.runes);

↳ (128512)

High
Surrogate
Low

Surrogate Pair Formula

$$\text{Code Point} = \frac{(\text{highSurrogate} - 0xD800) * 0x400}{0x10000} + \frac{(\text{lowSurrogate} - 0xDC00)}{0x10000}$$

$$\underline{0xD85B} \rightarrow \underline{16^3 \times 13} + \underline{16^2 \times 8} + \underline{16^1 \times 0} + \underline{16^0 \times 0}$$

A-10

B-11

C-12

D-13

$$53248 + 2048$$

$$= \underline{55296}$$

$$\underline{0x400} \rightarrow 16^2 \times 4 + 16^1 \times 0 + 16^0 \times 0$$

$$= \underline{1024}$$

$$\underline{0xDC00} \rightarrow 16^3 \times 13 + 16^2 \times 12 + 16^1 \times 0 + 16^0 \times 0$$
$$= 53248 + 3072$$

$$= \underline{56320}$$

$$\underline{0x10000} = 16^4 \times 1$$

$$= 65536$$

$$\text{Code Point} = [55357 - 55296] * 1024 + (56832 - 56320) + 65536$$

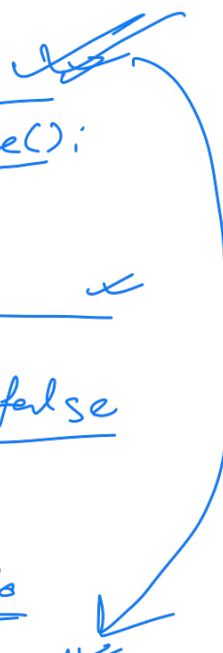
$$= 61 \times 1024 + 512 + 65536$$

$$= 62464 + 66048$$

$$= \underline{128512} \quad \checkmark$$

Strings are immutable :-

```
void main() {  
    String name = "Hello";  
    print(name); // Hello  
    print(name.hashCode);  
    String ucName = name.toUpperCase();  
    print(ucName); // HELLO  
    print(ucName.hashCode);  
    print(identical(name, ucName)); // false  
  
    String sameObj = name;  
    print(sameObj); // Hello  
    print(sameObj.hashCode);  
    print(identical(name, sameObj)); // true  
}
```



Escape Characters :

\n → New Line
\t → tab space
' → Single Quote
" → Double Quote
\\ → Backslash

```
void main() {  
    print("Hello\nDart");  
    // Hello  
    // Dart  
    print("Hello\tDart");  
    // Hello _____ Dart  
    print('I\'m learning Dart.');
```

// I'm learning Dart.

Raw String

```
void main() {  
    String raw = "Raw String: \n won't  
    create new Line";  
    print(raw);  
}
```

Properties & Methods :-

```
void main() {  
    String text = "Dart Programming";  
    print(text.length); // 16  
    print(text.isEmpty); // false  
    print(text.isNotEmpty); // true  
    print(text.runtimeType); // String  
    print(text[0]); // D  
    print(text[3]); // t  
    print(text.substring(0, 6));  
    print(text.substring(4)); // Dart  
                                // _Programming  
    print(text.contains("Dart"));  
                                // true  
    print(text.startsWith("Da")); // true  
    print(text.endsWith("ing")); // true  
}
```

